Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA (Remote)

PROFESSIONAL SUMMARY

Software engineer and applied AI builder who **owns complex integrations end-to-end** - the core of ElevenLabs' FDE role. Built AgentProbe from scratch: a **15-endpoint FastAPI platform** integrating multiple LLM providers (Anthropic, OpenAI, Groq, Ollama), a 3-agent LangGraph system with tool calls and inter-agent state handoffs, and a **5-metric evaluation engine** with CI/CD gates - the full stack for deploying agentic AI at scale. **8+ years production Python**; shipped 20+ full-stack modules across 120+ Agile sprints at Entrata (100K+ MAU, 1.6M records). Experienced translating complex technical requirements into working systems for stakeholders across engineering and business teams. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

Python (8+ years production) | API Integration & Software Architecture | Agentic Systems (LangGraph, LangChain)

Multi-Provider LLM Integration | FastAPI / REST Platform Layer | End-to-End Project Ownership

RAG Pipelines & Vector Search (ChromaDB) | LLM Evaluation & Observability | Docker / AWS / CI/CD

TypeScript / JavaScript / React | Enterprise-Scale Delivery (100K+ MAU) | Cross-Functional Technical Communication

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Took full ownership of **20+ full-stack modules** (Python, TypeScript/JavaScript, PostgreSQL) across **120+ Agile sprints** - delivered end-to-end from scoping to production for **100K+ monthly active users**, collaborating daily with engineers, product managers, and business counterparts.
- Applied **Python, PyTorch, and OpenAI API** to automate data transformation across **1.6M financial records** - production AI integration on regulated enterprise data with output validation and quality monitoring.
- Led Docker-based containerization for 30%+ of microservices, cutting execution time by **50%**; improved production query performance by **30%**; maintained **99.5% uptime** across enterprise SaaS for 6+ months.
- Migrated client data to **AWS S3** with Libsodium-encrypted pipelines - **100% PHI compliance, zero security incidents** across 100K+ users; built 5+ platform tools adopted across squads, reducing resolution time by **40%**.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Worked directly with 30+ students to solve complex technical problems - translating intricate concepts into actionable solutions, same communication pattern required when working with customer engineering teams at ElevenLabs.

PROJECTS

AgentProbe: AI Agent Integration Platform - LLMs, APIs, Agentic Systems OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

Built a complete enterprise-grade AI integration platform - exactly the kind of solution ElevenLabs FDEs build for strategic customers. **API integration**: multi-provider LLM routing across Anthropic, OpenAI, Groq, Ollama - dynamic selection by task, latency, and capability. **Agentic system**: 3-agent LangGraph system (Orchestrator, RAG Tracer, Auto Fix) with multi-step reasoning, tool calls, failure handling, and inter-agent state handoffs - the architecture ElevenAgents customers need for voice and chat agent deployments. **Platform layer**: 15-endpoint FastAPI backend + React observability dashboard - the shared integration surface any enterprise team connects to. **Evaluation**: 5-metric LLM-as-judge engine with CI/CD gates - reliability and observability built in from day one. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: Healthcare Enterprise Integration - LLMs + FHIR APIs

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings

End-to-end enterprise AI integration for a regulated domain - NLP routing natural language queries to FHIR APIs via LLMs, BERT, and embeddings. **HIPAA-compliant RBAC. ~60% retrieval time reduction.** Demonstrates the full-ownership integration pattern ElevenLabs FDEs execute with enterprise customers. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection → evaluation → deployment. **85%+ accuracy.**

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, Multi-Agent Orchestration, LLM APIs (Anthropic, OpenAI, Groq, Ollama), RAG Pipelines, ChromaDB, Vector Search, LLM-as-Judge Evaluation, CI/CD Eval Gates, Prompt Engineering, Context Engineering

Backend & Infra: FastAPI, Flask, React, PostgreSQL, MySQL, Docker, AWS (S3, EC2, RDS, Glue), Azure, CI/CD (GitHub Actions), RabbitMQ, Microservices

ML & Data: PyTorch, scikit-learn, BERT, spaCy, Hugging Face, NumPy, Pandas, Feature Engineering, Statistical Modeling

Process: TDD/BDD, Integration Testing, Open-Source Development, Git, JIRA, Agile/Scrum, Claude Code, GitHub Copilot